Course code: MA2005-1

Course Tittle: Linear Algebra and its Application

Model MCQ

	UNIT I	
1.	Rank of an identity matrix of order 4 is_____	
	a) 4	b) 3
	c) 1	d) 0
2.	System of equations AX=B has solutions if	
	a) Rank[A] = number of unknowns	c) Rank[A] $\neq$ number of unknowns
	b)Rank[A] = Rank[A $\vert$ B]$\leq$ number of unknowns	d) Rank[A] $\neq$ Rank[A $\vert$ B]
3.	System AX=0 has nontrivial solution if and only if	
	a) Determinant of A=Determinant of X	b) Determinant of A =0
	c) Rank[A] $\neq$ Rank[A $\vert$ B]	d) Determinant of A $\neq$ 0
4.	Which of the following statements is incorrect?	
	a) Rank of a matrix A is the number of nonzero rows in its echelon form	b) Rank of a matrix A is equal to number of pivot columns in it.
	c) Rank of a matrix A is number of free variables in the system AX=B	d) Rank of a matrix A is number of basic variables in the system AX=B
5.	If $[A:B] = \begin{bmatrix} 2 & 3 & -1 & : & 5 \\ 0 & 1 & 3 & : & -1 \\ 0 & 0 & \frac{k}{3}-6 & : & 5 \end{bmatrix}$ then the value of k for which the system is inconsistent is_____	
	a) 18	b) -18
	c) 6	d) -6
6.	System of equations AX=B has infinite number of solutions if	
	a) Rank[A] = Rank[A $\vert$ B] = number of unknowns	b) Rank[A] = Rank[A $\vert$ B] < number of unknowns
	c) Rank[A] $\neq$ Rank[A $\vert$ B]	d) Rank[A] = Rank[A $\vert$ B] > number of unknowns
7.	If $A = \begin{pmatrix} -1 & 1 \\ 2 & 4 \end{pmatrix}$ then $tr(AA^T)$ is	
	a) 21	b) 22
	c) 3	d) 26
8.	In the LU factorization of $A = \begin{bmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{bmatrix}$, the matrix U is	
	a) $A = \begin{bmatrix} 3 & 7 & 2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix}$	b) $A = \begin{bmatrix} 3 & -7 & -2 \\ 0 & -2 & 2 \\ 0 & 0 & -1 \end{bmatrix}$
	c) $A = \begin{bmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix}$	d) $A = \begin{bmatrix} 3 & -7 & -2 \\ 0 & 5 & 1 \\ 0 & 0 & 0 \end{bmatrix}$
9.	The eigen values of $A = \begin{bmatrix} 5 & 6 & 17 \\ 0 & -19 & 23 \\ 0 & 0 & 37 \end{bmatrix}$ are	
	a) -19, 37, 5	b) $2, -3, 7$

	c) 5, 19, 37	d) 0, 5 ,17
10.	The eigen values of $A = \begin{bmatrix} 2 & 4 & -6 \\ 0 & 3 & 6 \\ 0 & 0 & 3 \end{bmatrix}$ are	
	a) 2,2,4	b) 2,3,2
	c) 1,3,4	d) 2,3,3
11.	A $n \times n$ matrix A is diagonalizable if and only if	
	a) A has n distinct eigen values	b) A has repeated eigen vectors
	c) A has repeated eigen values	d) A has linearly independent eigen vectors
12.	If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & 2 \\ -1 & 3 \end{bmatrix}$ then $tr(B^{-1}AB)$ is	
	a) 4	b) 5
	c) 10	d) 8
13.	If $A = \begin{bmatrix} -1 & 2 \\ 3 & -3 \end{bmatrix}$ then $tr(AA^T)$ is	
	a) 23	b) 10
	c) 1	d) 22
14.	Eigen value of a triangular matrix A are	
	a) diagonal element of A	b)sum of diagonal element of A
	c) Determinant of A	d) Trace of A
15.	Eigen values of A= $\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$ are	
	a) 3,-3	b) -3,-5
	c) 3,5	d) 5,0
16.	Let $A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$, the eigen value $\lambda_1 = 2, \lambda_2 = 3$ and $\lambda_3 =$ _____	
	a) 3	b) 2
	c) 5	d) 7
17.	The eigen vectors of the matrix A is the nontrivial solution of the equation _____	
	a) $AX = 0$	b) $AX = B$
	c) $AX = \lambda X$	d) none
18.	If $A = \begin{bmatrix} 1 & 3 & 2 \\ 5 & 6 & -1 \\ 2 & -2 & 3 \end{bmatrix}$ $B = \begin{bmatrix} -6 & 2 & 5 \\ 2 & 1 & 8 \\ 4 & -2 & 2 \end{bmatrix}$ then tr(A-B) is	
	a) 1	b) 12
	c) 13	d) 7
19.	If A is a square matrix, then det (A)=	
	a) sum of the eigen values of A	b) product of eigen values of A
	c) product of diagonal elements of A	d) sum of diagonal elements of A
20.	The sum of the eigen values of $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & -1 & -1 \end{bmatrix}$ is	
	a) -2	b) 3
	c) 6	d) 5

21.	Let $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, then eigen values of A are $\lambda_1 = 1, \lambda_2 = 3$ and $\lambda_3 =$ _____	
	a) -2	b) 1
	c) 2	d) -1
22.	Let A be an $n \times n$ matrix, then A is diagonalizable if and only if	
	a) A has linearly dependent eigen vectors.	b) A has n linearly independent columns.
	c) A has n linearly independent eigen vectors.	d) A has (n-1) linearly independent eigen vectors.
23.	For which value of k the system of two equations $x + y = 1$ $2x + 2y = k$ has infinite solutions:	
	a) $k = 0$	b) $k = 1$
	c) $k = 2$	d) any value of $k \neq 1$
24.	The eigen values of $A = \begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix}$ are	
	a) 7,4	b) **9,2**
	c) 8,3	d) 6,5
25.	Spectrum of the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ is:	
	a) $\{3, 2, 1\}$	b) $\{1, 3, 1\}$
	c) $\{0, 0, 6\}$	d) $\{2, 3, 0\}$
26.	Let $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$, then $tr(A + B)$ is	
	a)19	b) 17
	c) 20	d) 21
27.	Which of the following statements are/is correct for the system $AX = B$, $when\ A\ is\ a$ 3×4 matrix of rank 2. I. Matrix A has 2 non-pivot columns. II.System $AX = B$ has 2 basic variables.	
	a) Only I is correct	b) Only II is correct
	c) Both are correct	d) Both are not true
28.	If A is 3×4 matrix the maximum possible value of the rank A is	
	a) 4	b) **3**
	c) 2	d) 0
29.	If a matrix A is reduced to row echelon form, then rank of the matrix is	
	a) number of zero rows	b) **number of non-zero rows**
	c) total number of rows	d) total number of columns
30.	The characteristic equation of the matrix $A = \begin{bmatrix} 2 & 0 \\ 4 & 7 \end{bmatrix}$ is:	
	a) $\lambda^2 - 9\lambda + 14 = 0$	b) $\lambda^2 - 14\lambda + 9 = 0$
	c) $\lambda^2 - 9\lambda - 14 = 0$	d) $\lambda^2 - 14\lambda - 9 = 0$
31.	If $3x + 2y + z = 0, x + 4y + z = 0, 2x + y + 4z = 0$,be a system of equations, then	
	a) it is inconsistent	b) **it has only the trivial solution $x = 0, y = 0, z = 0$**
	c)	d)

	it can be reduced to a single equation and so a solution does not exist.	determinant of the matrix of coefficient is zero.
32.	Which of the following statement is correct I. Trace of the square matrix is equal to product of eigenvalues II. Trace of the square matrix is equal to sum of diagonal entries of the matrix.	
	a) only I is true	b) only II is true
	c) both I and II are true	d) neither I nor II is true.
33.	If the LU factorization of a matrix A is A=LU where A=$\begin{bmatrix} 2 & 5 \\ -3 & -4 \end{bmatrix}$ then U is	
	a) $\begin{bmatrix} 2 & 5 \\ 0 & 7 \end{bmatrix}$	b) $\begin{bmatrix} 1 & 4 \\ -5 & 1 \end{bmatrix}$
	c) $\begin{bmatrix} 2 & 5 \\ 1 & 7 \end{bmatrix}$	d) $\begin{bmatrix} 1 & 4 \\ -7/2 & 1 \end{bmatrix}$
34.	Which of the following method is not used to find the solution of the system of equations:	
	a) Gauss elimination method	b) Gauss seidel method
	c) Rayleigh's power method	d) LU decomposition method
35.	Which of the following is not a valid elementary operation?	
	a) $R_i \leftrightarrow R_j$	b) $R_i \to R_j + kR_i$
	c) $R_i \to kR_i$	d) $R_i \to 1 + kR_i$
36.	The value of h for which the augmented matrix $\begin{bmatrix} 1 & -3 & -2 \\ 5 & h & -7 \end{bmatrix}$ of a linear system is inconsistent	
	a) $h = -15$	b)$h \neq -15$
	c) $h = 5$	**d)** $h = -1$
37.	Number of pivot columns in the matrix $\begin{bmatrix} 0 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$ is:	
	a) 1	b) 2
	c) 3	d) 4
38.	Which of the following vector is an eigen vector of the matrix $\begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$:	
	a) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	b) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$
	c) $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$	d) $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
	UNIT II	
39.	A subset S of vector space V over field F is a basis of V if S is linearly independent and	
	$a)\ Span(S) = F$	$b)\ Span(V) = S$
	$c)\ Span(S) = V$	d) $Span(S) = S$
40.	If $\{u_1, u_2, \ldots\ldots\ldots, u_n\}$is linearly independent set, then $c_1u_1 + c_2u_2 + \ldots\ldots + c_nu_n = 0$ has	
	a)infinitely many solutions	b) unique nonzero solution
	c) unique zero solution	d) both B and C are possible
41.	The row space of an $m \times n$ matrix A is a subspace of	
	a) R^n	b) A
	c) R^m	d) R
42.	In two dimensional Euclidean space identity element with respect to vector addition is	
	a) (1,1)	b) (0,0)
	c) (-1,-1)	d) doesn't exists

43.	In three dimensional Euclidean space inverse of (-2,-2,0) with respect to vector addition is	
	a) (0,0,0)	b) (2,2,2)
	c) (2,2,-2)	d) (2,2,0)
44.	Let $T: R^4 \to R^4$be the linear transformation with null space of T={(0,0,0,0)} . Then nullity of T is	
	a) 0	b) 1
	c) 4	d) 2
45.	The dimension of the subspace of R^3 spanned by $\{(2,0,1),(5,2,1),(1,2,-1)\}$ is	
	a) 3	b) 2
	c) 1	d) 0
46.	The column space of an $m \times n$ matrix is a subspace of	
	a) R^m	b) R^n
	$c)$ R^{mn}	d) R^{m+n}
47.	Let $T: R^3 \to R^3$ be the linear transformation with nullity of T is 1 then rank of T is	
	a)1	b) 2
	c) 3	d) 4
48.	If $T: V \to W$ is a linear transformation, which of the following statements is not true?	
	a) $T(xy) = T(x)T(y) \forall x, y \in V$	b) $T(-x) = -T(x) \forall x \in V$
	c) $T(0) = 0 \; for \; 0 \in V$	d) $T(2x) = 2T(x) \forall x \in V$
49.	The set $W = \{(a,0,0): a, b \; \epsilon R\}$ is a subspace of .	
	a) $R^2(R)$	b) $R(R)$
	c) $R^3(R)$	d) $R^4(R)$
50.	If $T: R^2 \to R^3$ is a linear transformation such that $T(1,0) = (2,3,1)$, $T(1,1) = (3,0,2)$ then which of the following is true?	
	a) $T(x,y) = (x+y, 2x+y, 3x-3y)$	b) $T(x,y) = (2x+y, 3x-3y, x+y)$
	c) $T(x,y) = (2x-y, 3x+3y, x-y)$	d) $T(x,y) = (x-y, 2x-y, 3x+3y)$
51.	If $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, v = \begin{bmatrix} -5 \\ 7 \end{bmatrix}$ then inner product of $u$and v is	
	a) 9	b) 19
	c) -3	d) 3
52.	A system of equation $AX = b$ has solution if and only if	
	a) $b \in Col(A)$	b) $b \in Row(A)$
	c) $b \in Null(A)$	d) $b \in Col(A)$ and $b \in Row(A)$
53.	Let $T: R^2 \to R^2$ be the linear transformation defined by $T(x,y) = (x,0)$, then the matrix of linear transformation with respect to standard basis is	
	a) $\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$	b) $\begin{bmatrix} 0 & -1 \\ 0 & 1 \end{bmatrix}$
	c) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	d) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
54.	Which of the following statement is not true if $T: U \to V$is a linear transformation	
	a) Range space of T is a subspace of V	b) Null space of T is a subspace of U
	c) V is always the range of T	d) None
55.	Two vectors u and v in R^n are orthogonal if	
	a) $u.v = v.u$	b) $u.v = 1$

	c) $u.v \geq 0$	d) $\boldsymbol{u.v = 0}$
56.	If $u = \begin{bmatrix} 2 \\ 3 \\ -5 \end{bmatrix}, v = \begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix}$ then inner product of u and v is	
	a) -5	b) -11
	c) 19	d) 11
57.	Which of the following set of vector is not orthogonal	
	$a)\ \left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}$	b) $\left\{\begin{bmatrix} -2 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$
	$c)\ \left\{\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$	d) $\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$
58.	If a basis of a vector space has 3 elements, then dimension of the vector space is_____	
	a) less than 4	b) greater than 3
	c) **3**	d) 0
59.	If $v = [1 \quad 1 \quad 1 \quad -1]^T$ then normalized form of v is	
	a) $\left[\frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \quad \frac{-1}{\sqrt{2}}\right]^T$	b) $\left[\frac{1}{\sqrt{3}} \quad \frac{1}{\sqrt{3}} \quad \frac{1}{\sqrt{3}} \quad \frac{-1}{\sqrt{3}}\right]^T$
	c) $\left[\frac{1}{2} \quad \frac{1}{2} \quad \frac{1}{2} \quad \frac{-1}{2}\right]^T$	d) $[1 \quad 1 \quad 1 \quad 1]^T$
60.	The coordinate of $v = (1,2,4)$ relative to the standard basis of R^3 is	
	$a)$ (1,2,0)	$b)$ (1,0,0)
	$c)$ (1,2,4)	d) (0,2,4)
61.	Let $T: R^2 \to R^2$ be the linear transformation defined by $T(x, y) = (x, 0)$. The null space of T is	
	a) {(x, 0)}	b) **{(0, y)}**
	c) {(0, x)}	d) {(0, 0)}
62.	The set of all solution to the homogeneous equation system $AX = 0$, where A be an $m \times n$ matrix is	
	a) Null space of A	b) Column space of A
	c) Row space of A	d) Null space of A and column space of A
63.	Which of the following are linear combination of u= (0,-2,2) and v=(1,3,-1)?	
	a)(1,0,0)	b)(2,2,2)
	c) both (a) &(b)	d) neither (a) nor (b)
64.	Let $T: R^2 \to R^2$ be the linear transformation defined by $T(x, y) = (0, y)$.Then matrix of linear transformation with respect to standard basis is	
	a) $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$	b) $\begin{bmatrix} 0 & -1 \\ 0 & -1 \end{bmatrix}$
	c) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	d) $\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$
65.	If $S = \{v_1, v_2, ...v_n\}$ is a set of vectors in a finite dimension vector space V, then S is called a basis for V if	
	a) S spans V	b) S is linearly independent
	c) either (a) or (b)	d) both (a) and (b)
66.	Let $T: R^4 \to R^4$ be the linear transformation defined by $T(x_1, x_2, x_3, x_4) = (x_2, x_3, 0,0)$. Then null space of T consist of vectors of the form	
	a) $(0,0, x_2, x_3)$	b) $(x_1, 0,0,0)$
	c) $(x_1, 0,0, x_4)$	d) $(x_1, x_1, x_3, 0)$
67.	If the inner product of two vectors is zero then they are called	
	a) orthogonal vectors	b) null vectors
	b) unit vectors	d) dependent vectors

68.	The matrix of linear transformation $T(x,y)=(y,x)$ with respect to the standard basis is	
	a) $\begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$	b) $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$
	c) $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$	d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
	UNIT III	
69.	The quadratic form $3x_1^2 - 4x_1x_2 + 6x_2^2$ is	
	a) **Positive definite**	b) Positive semi definite
	c) Negative definite	d) Neither negative definite nor positive definite
70.	In QR factorization of matrix A, Q is a matrix whose columns form an orthonormal basis for	
	a) Col A	b) Null(A)
	c) Row(A)	d) both Null(A) and Row(A)
71.	The matrix of the quadratic form $Q(X) = 5{x_1}^2 + 5{x_2}^2 - 4x_1x_2$ is	
	a) $\begin{bmatrix} 5 & -2 \\ -2 & 5 \end{bmatrix}$	b) $\begin{bmatrix} 5 & 2 \\ 2 & 5 \end{bmatrix}$
	c) $\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$	d) $\begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$
72.	Matrix A of the quadratic form $Q(x) = {x_1}^2 - 8x_1x_2 - 5{x_2}^2$ is	
	a) $A = \begin{pmatrix} 1 & -8 \\ -8 & -5 \end{pmatrix}$	b) $A = \begin{pmatrix} 1 & -4 \\ -4 & -5 \end{pmatrix}$
	c) $A = \begin{pmatrix} 0 & -8 \\ -8 & 0 \end{pmatrix}$	d) $A = \begin{pmatrix} 1 & -4 \\ -5 & -4 \end{pmatrix}$
73.	Let B be the mean deviation form of N observation vector then covariance matrix is	
	$a)\ S = \frac{1}{N-1}B^T$	b) $S = \frac{1}{N-1}BB^T$
	c) $S = \frac{1}{N+1}BB^T$	d) $S = \frac{1}{N}BB^T$
74.	Singular values of $A = \begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}$ are	
	a) 1,-3	b) 9,1
	$c)\ \sqrt{3}$,1	d) 3,1
75.	The quadratic form $3x^2 + 5y^2 + z^2$ is	
	a) Positive definite	b) Negative definite
	c) Negative semi definite	d) Positive semi definite
76.	In PCA, what are the principal components	
	a) Features of the data set	b) Eigen values of the covariance matrix
	c) **Eigen vectors**	d) Data points in the data set
77.	In PCA, what is the relationship between the first principal component and the second principal component	
	a) **They are orthogonal to each other**	b) They are positively correlated
	c) They are negatively correlated	d) There is no defined relationship between them
78.	The matrix of the quadratic form $4x_1x_2 + 6x_1x_3 - 8x_2x_3$	

	a) $\begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & -4 \\ 2 & -4 & 0 \end{bmatrix}$	b) $\begin{bmatrix} 0 & 1 & 3 \\ 2 & 0 & -4 \\ 3 & -4 & 0 \end{bmatrix}$
	c) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & -4 \\ 3 & -4 & 1 \end{bmatrix}$	d) $\begin{bmatrix} 0 & 2 & 3 \\ 2 & 0 & -4 \\ 3 & -4 & 0 \end{bmatrix}$
79.	A quadratic form Q(x) is semi positive definite if	
	a)$Q(x) > 0$ for all $x \neq 0$	b) $Q(x) \geq 0$ for all x
	c) $Q(x) < 0$ for all $x \neq 0$	d) $Q(x) \leq 0$ for all x
80.	If $A = \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}$ and $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, then $x^T Ax$ is	
	a) $4{x_1}^2 + 3{x_2}^2$	b) $2{x_1}^2 + 3{x_2}^2 + 7x_1x_2$
	c)$2{x_1}^2 + 3{x_2}^2 - 7x_1x_2$	d) $2{x_1}^2 + 3{x_2}^2$